



Connor Parrow

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EDUCATION

University of Notre Dame	South Bend, IN
M.S., Applied and Computational Mathematics,	January 2025
Ph.D., Applied and Computational Mathematics,	Expected May 2027
Hobart and William Smith Colleges	Geneva, NY
B.S., Mathematics and Biology; GPA: 3.96, <i>summa cum laude</i>	May 2021

RESEARCH INTERESTS

Numerical Analysis of PDEs, Multiphysics Systems, Partitioned & Operator-Splitting Methods, Fluid-Structure/Poroelastic Interaction.

PUBLICATIONS

- Parrow, C., Mangini, A., Bukač, M. "A partitioned, second-order method for two-phase flow." *Submitted to Advances in Computational Mathematics* (July 2026).
- Parrow, C., Bukač, M. "Stability and Error Analysis of a Partitioned Robin-Robin Method for Fluid Poroelastic Structure Interactions." *Journal of Scientific Computing* 106.2 (2026):40. <https://doi.org/10.1007/s10915-025-03143-1>
- Parrow, C., Bukač, M. "A Robin–Robin strongly coupled partitioned method for fluid–poroelastic structure interaction." *Journal of Numerical Mathematics* 33.3 (2025): 289-312. <https://doi.org/10.1515/inma-2024-0057>
- Bell, J.R., Nwobbi, L., Parrow, C., Wheeler, A., & Zelleke, E. "Graceful wind chimes." *Involve, a Journal of Mathematics* 14.4 (2021): 683-696.

AWARDS & HONORS

- Advance Research Award Honorable Mention Award April 2026
- Graduate School Professional Development Award March 2025
- Graduate Student Government Professional Development Award April 2025
- ACMS Department Travel Award June 2025, January 2024
- SIAM AN24 Travel Award July 2024
- SIAM EDGE Participant Fellow July 2022
- Gilman Scholar Fall 2019

TEACHING & MENTORSHIP EXPERIENCE

University of Notre Dame	South Bend, IN
Instructor of Record, ACMS 20340 - Statistics for Life Sciences	Spring 2026
● Helped a class of 30 students learn to critically evaluate information and apply logical analysis in order to make proper decisions on statistical methods to use in life and health sciences problems. ● Taught the importance of analytical models such as discrete and continuous probability distributions, sampling distributions, confidence intervals, hypothesis testing and other statistical methods.	
Undergraduate Research Project Mentor	January 2025-May 2026
● Guided two undergraduate students in developing simulation algorithms for the Cahn-Hilliard Navier-Stokes equations to study multiphase fluid dynamics in numerical experiments.	
Teaching Assistant, Applied and Computational Mathematics and Statistics	August 2022 – Present
● Taught programming labs in Python and RStudio, mentoring students in numerical modeling and data analysis.	
EDGE Foundation	Knoxville, TN
Program Mentor	June 2024 – July 2024
● Organized problem sessions for and mentored 20 incoming Ph.D. students in mathematics across the country, covering topics such as Abstract Algebra, Real Analysis, and Measure Theory.	



TALKS & PRESENTATIONS

Invited Presentations

- “Recent Advances in Partitioned Numerical Methods for Fluid-Poroelastic Structure Interaction”, Minisymposium on Recent Advances in Galerkin Methods for Physical Systems, **SIAM Conference on Computational Science and Engineering** February 2027 (**Upcoming**)
- “On Robin-Based Partitioned Schemes for Fluid-Poroelastic Structure Interaction: Part 2”, Special Session on Theoretical and Numerical Methods for PDEs related to Fluids and Fluid-Solid Interactions, **Joint Math Meetings** January 2027 (**Upcoming**)
- “Partitioned Methods for Fluid-Poroelastic Structure Interactions”, **Association For Women in Science STEM Luncheon** November 2025
- “Mathematics behind Vaccine Policies”, **Science Policy Journal Club** November 2025
- “A Second Order Partitioned Method for Fluid-Poroelastic Structure Interaction”, Minisymposium on Recent Advances in High-Order Methods for Non-linear Multiphysics Systems, **International Conference on Spectral and High Order Methods** July 2025
- “Partitioned Methods for Fluid-Poroelastic Structure Interaction Based on Robin Boundary Conditions”, Minisymposium on The Finite Element Method for Problems in Multiphysics and Geometry: Part 1, **30th Biennial Numerical Analysis Conference** June 2025
- “Backward Euler-Forward Euler Partitioned Methods for FPSI”, Research by Women of EDGE Special Session, **Joint Math Meetings** January 2025

Contributed Talks & Posters

- “A Fully Partitioned Robin-Robin Method for Fluid-Poroelastic Structure Interaction: Stability and Improved Convergence”, **Midwest Numerical Analysis Day** April 2026
- “Bridging Fluid and Structure: A Fast, Accurate Method for Stokes–Biot Interaction”, **COSE Research Horizons Symposium** December 2025
- “Second Order Partitioned Methods for Fluid-Poroelastics Interactions”, **Graduate Interdisciplinary Research Symposium** April 2025
- “A Second-Order Partitioned Method for Fluid-Poroelastic Structure Interaction”, Minisymposium: AWM Workshop, **SIAM AN24** July 2024
- “Refactorization of Cauchy’s method: a second-order partitioned method for fluid-poroelastic material interaction”, **Mathematical Models and Numerical Methods for Multiphysics Systems Conference** April 2024

LEADERSHIP & SERVICE

Association for Women in Mathematics (AWM), Student Chapter President August 2024 – Present

- Plan workshops and outreach programs promoting women in mathematics including over 20 members.

Graduate Student/PostDoc Seminar, Cofounder and Co-organizer February 2025 – Present

- Organize biweekly seminars for graduate students and postdocs to prepare for giving talks at conferences.

Society for Industrial and Applied Mathematics (SIAM), Chapter President May 2025 – Present

Society for Industrial and Applied Mathematics (SIAM), Chapter Treasurer August 2024 – May 2025

- Work with board members to plan events involving community outreach and career development.

Graduate Student Government, ACMS Department Representative August 2023 – May 2024

- Represented student interests on the Quality of Life subcommittee and advocated for coverage of dependents on graduate students health insurance.

TECHNICAL SKILLS

- **Modeling & Simulation:** Finite Element Methods, Multiphysics Simulation, CFD, Multiphase flow
- **Programming & Tools:** MATLAB, Python, C++, FreeFEM++, RStudio, LaTeX, Julia
- **Domain Expertise:** Fluid Dynamics, Fluid–Structure Interaction, Numerical Analysis, PDEs, Machine Learning